

VIDYA JYOTHI INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF INFORMATION TECHNOLOGY

LAB PROJECT REPORT



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Project Title	:	Online Examination System
Course Name	:	Object Oriented Programming through Java Lab
Course Code	:	A224592
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Year and Semester	:	II B.Tech II Semester



CERTIFICATE

This is to certify that the Object Oriented Programming through Java Lab project report entitled “ **ONLINE EXAMINATION SYSTEM** ” submitted by **D.Srinadh Goud (25915A1207)** to Vidya Jyothi Institute of Technology (An Autonomous Institution), Hyderabad, of B. Tech. II year, II Semester in Information Technology a Bonafide record of Lab project work carried out by them under my supervision.

Signature of Faculty

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Signature of Head of the Department

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Professor

Abstract

- The Online Examination System is a Java-based desktop application developed using Java Swing and MySQL database. The project is designed to automate the process of conducting objective-type examinations.
- Users can securely login into the system, attend multiple-choice questions, and receive automatic evaluation with score and percentage generation. The system provides a timer feature using multithreading, allowing the exam to end automatically when the time is completed.
- The application uses JDBC connectivity for communication with the MySQL database. The project reduces manual effort, improves efficiency, and provides fast and accurate result generation.
- Technologies used include Java, Swing, JDBC, MySQL, and Multithreading.

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Chapter 1 – Introduction

1.1 Project Overview

- The Online Examination System is used to conduct online tests using multiple-choice questions. The application allows users to login, attempt questions, and view results instantly.
- The system leverages multithreading to automatically terminate the examination when the time limits expire.
- Answers are evaluated instantly by the application to compute final test scores and percentages accurately.

1.2 Objectives

- To automate the examination process
- To reduce manual errors
- To generate results automatically
- To improve efficiency and accuracy

1.3 Scope of the Project

- The system can be used by educational institutions to conduct exams efficiently. Future enhancements can include cloud integration and online accessibility.

Chapter 2 – System Analysis

2.1 Functional Requirements:

- User Login
- Display Questions
- Automatic Evaluation
- Score Generation
- Timer using Threads

2.2 Non-Functional Requirements

- Security
- Performance
- Reliability
- User-Friendly Interface

2.3 Software Requirements

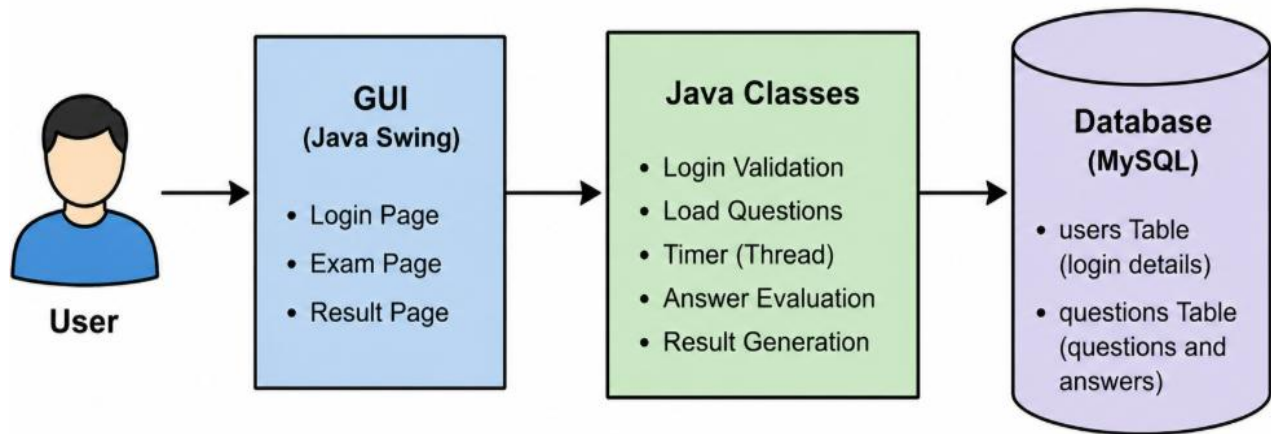
- Operating System: Windows
- Programming Language: Java
- IDE: VS Code / NetBeans
- Database: MySQL
- JDK: Version 8 or above

2.4 Hardware Requirement

- RAM 4 GB
- Processor Intel i3 or
above
- Storage 500GB

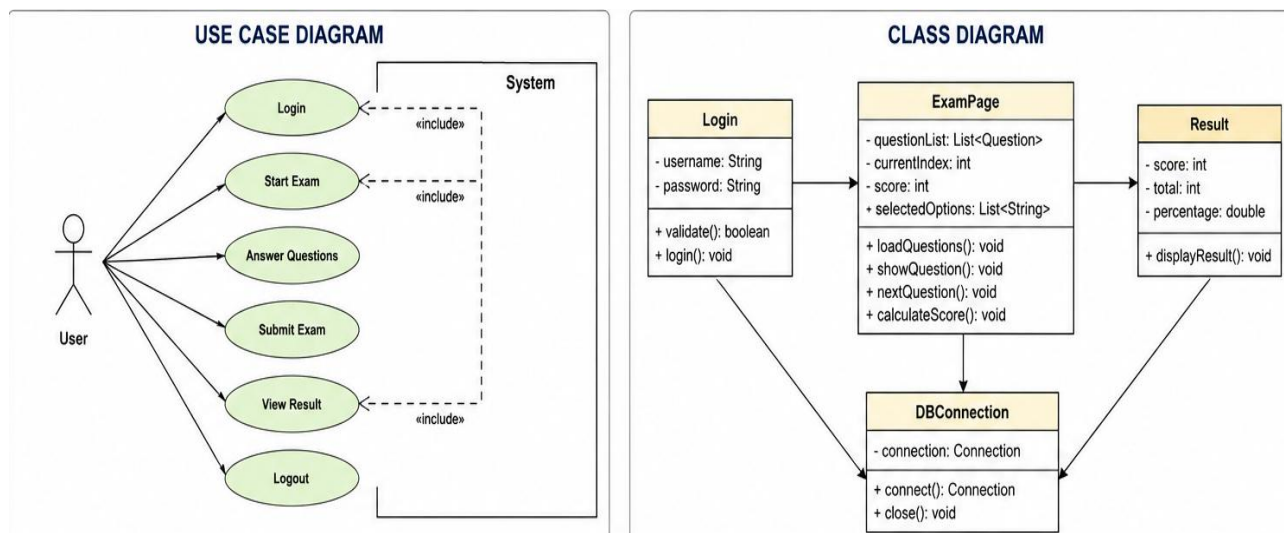
Chapter 3 – System Design

3.1 Block Diagram:

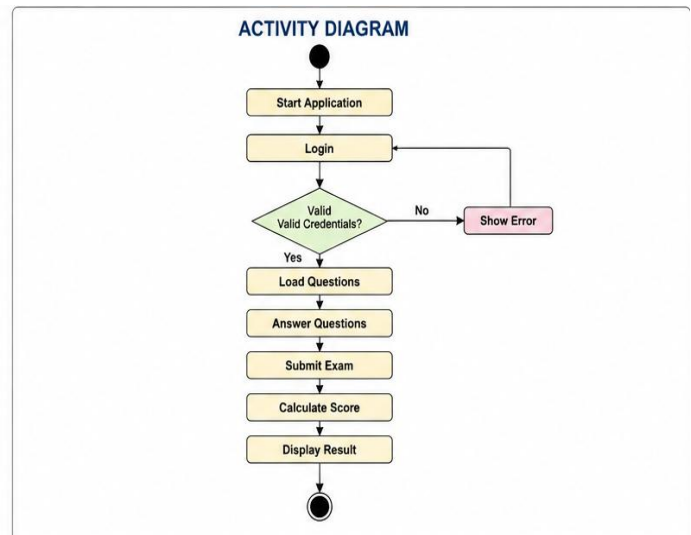
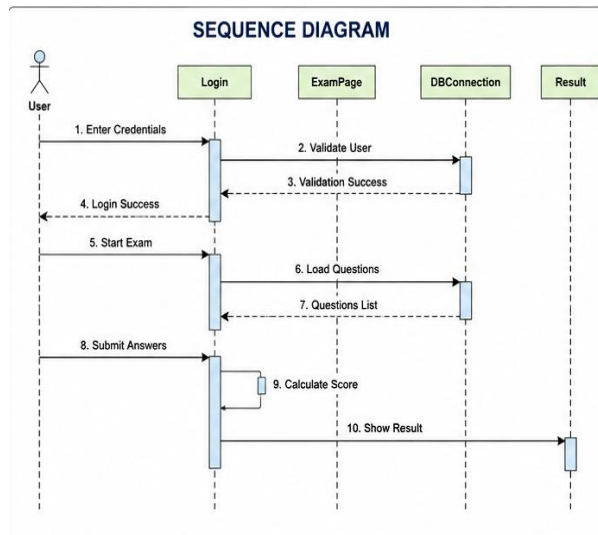


The Online Examination System uses a simple architecture where the user interacts with the Java Swing GUI. The GUI communicates with Java classes for processing login, questions, timer, and results. Java classes connect with the MySQL database using JDBC.

3.2 UML Diagram:



- User → GUI → Java Classes → Database



Use Case Diagram

Shows interactions between user and system such as:

- Login
- Attend Exam
- Submit Answers
- View Result

Class Diagram

Represents classes used in the project:

- Login
- ExamPage
- Result
- DBConnection

Sequence Diagram

User Login → Question Display → Evaluation → Result

Activity Diagram

Shows the workflow of the system from login to result generation.

Chapter 4 – Implementation

4.1 Modules Description

Module	Description
Login Module	Handles user authentication using username and password.
Questions	Displays questions and records user answers.
Report Module	Calculates and displays the final exam score.

4.2 Important Java Concepts Used

Example:

Mention syllabus concepts used:

- Classes & Objects
- Exception Handling
- Multithreading
- Collections
- File Handling
- AWT/Swing
- JDBC

4.3 Code Snippets

DataBase Connection Code:

```
import java.sql.Connection;
import java.sql.DriverManager;
public class DBConnection {
    static Connection con;
    public static Connection getConnection() {
        try {

            Class.forName("com.mysql.cj.jdbc.Driver");
```

```
        con = DriverManager.getConnection(  
            "jdbc:mysql://localhost:3306/online_exam",  
            "root",  
            "root"  
        );  
    } catch (Exception e) {  
        System.out.println(e);  
    } return con;  
}  
}
```

Login Code:

```
Connection con = DBConnection.getConnection();
```

Question Code:

```
Connection con = DBConnection.getConnection();
```

Result Code:

```
Connection con = DBConnection.getConnection();
```

Chapter 5 – Testing

5.1 Test cases :

Test Case ID	Input	Expected Output	Result
TC01	Valid Login	Login Success	Pass
TC02	Invalid Password	Error Message	Pass
TC03	Empty Username	Username Required Message	Pass
TC04	Timer Ends	Exam Auto Submitted	Pass
TC05	View Result	Result Displayed Successfully	Pass

Error Handling:

- **NumberFormatException** – Handles invalid numeric input entered by the user.
- **SQLException** – Handles database connection and query execution errors.
- **FileNotFoundException** – Handles missing or unavailable file errors.
- **NullPointerException** – Handles null value access errors in the program.
- **IOException** – Handles input and output related errors.

Chapter 6 – Results and Screenshots

Result:

The Online Examination System project was successfully developed using Java and MySQL. The system performs user login, question management, timer-based examination, and result generation efficiently.

It provides a simple and user-friendly interface for conducting online exams with accurate score calculation and database management.

Screenshots:

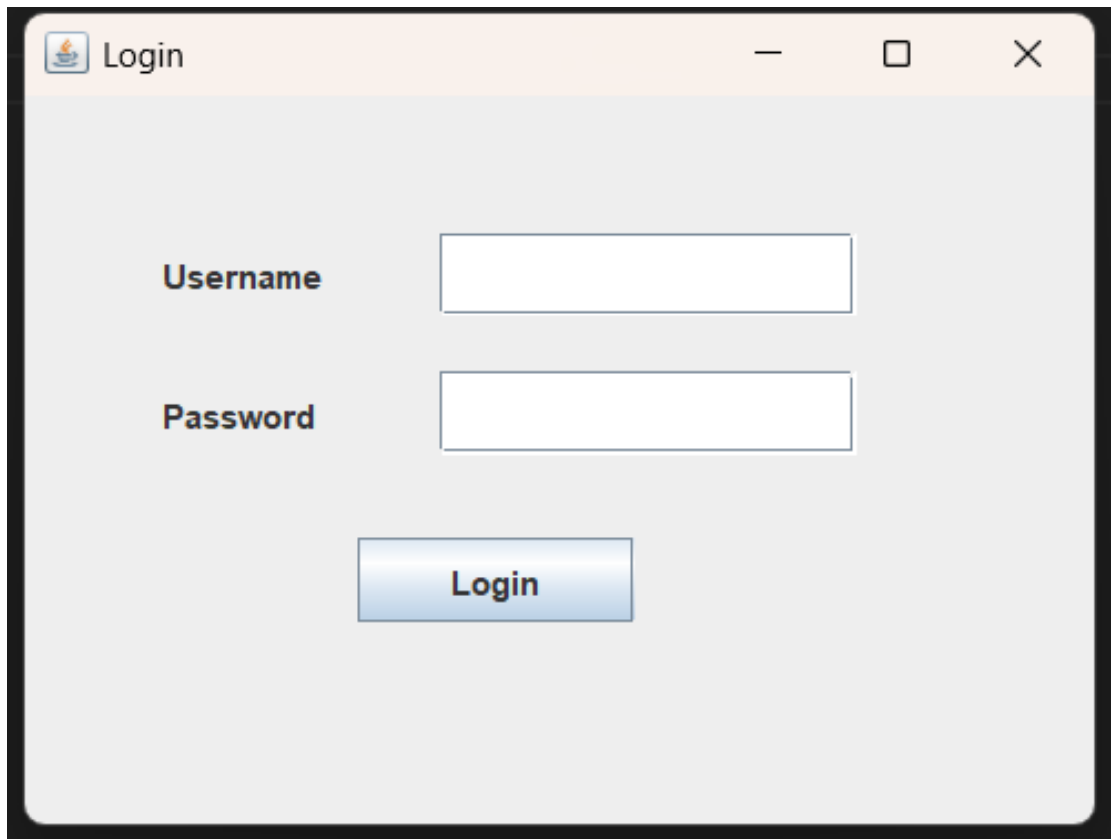


Figure 6.1: Login Screen

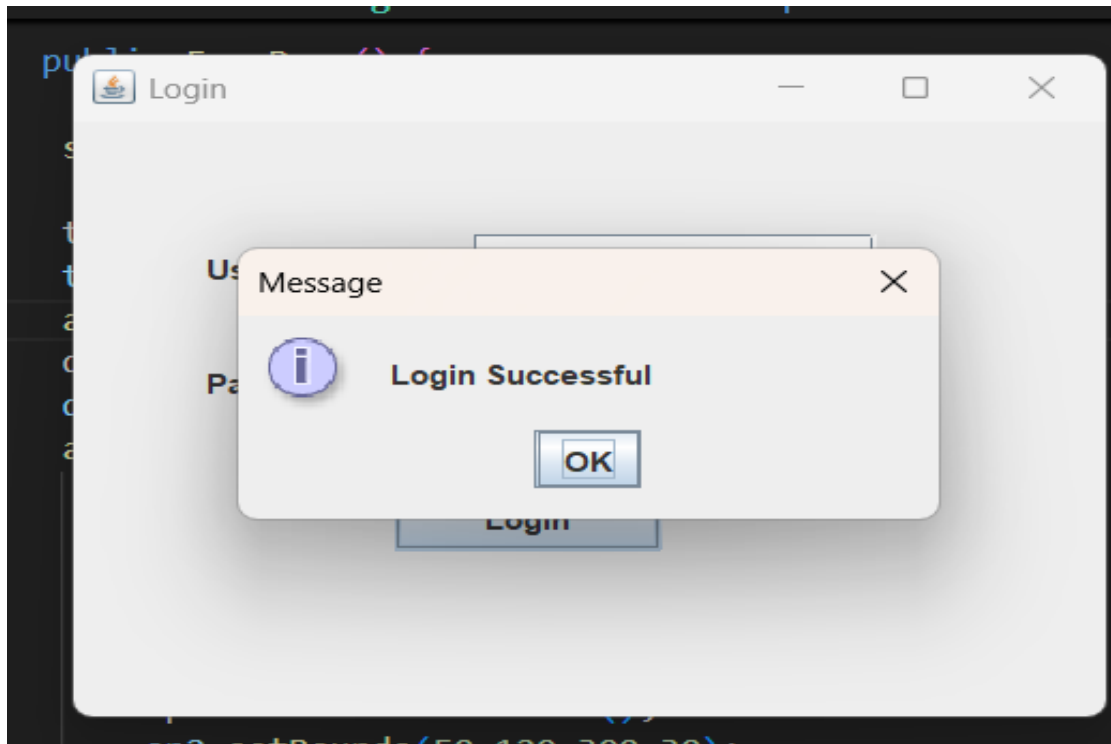


Figure 6.2: Authentication Screen

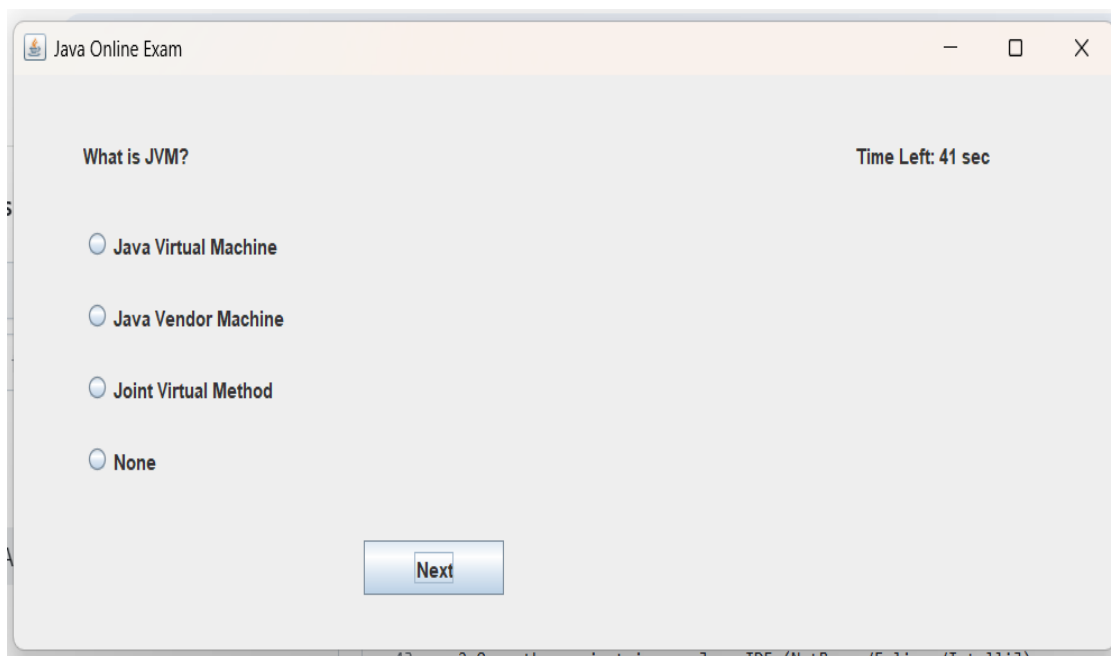


Figure 6.3: Online Examination Interface

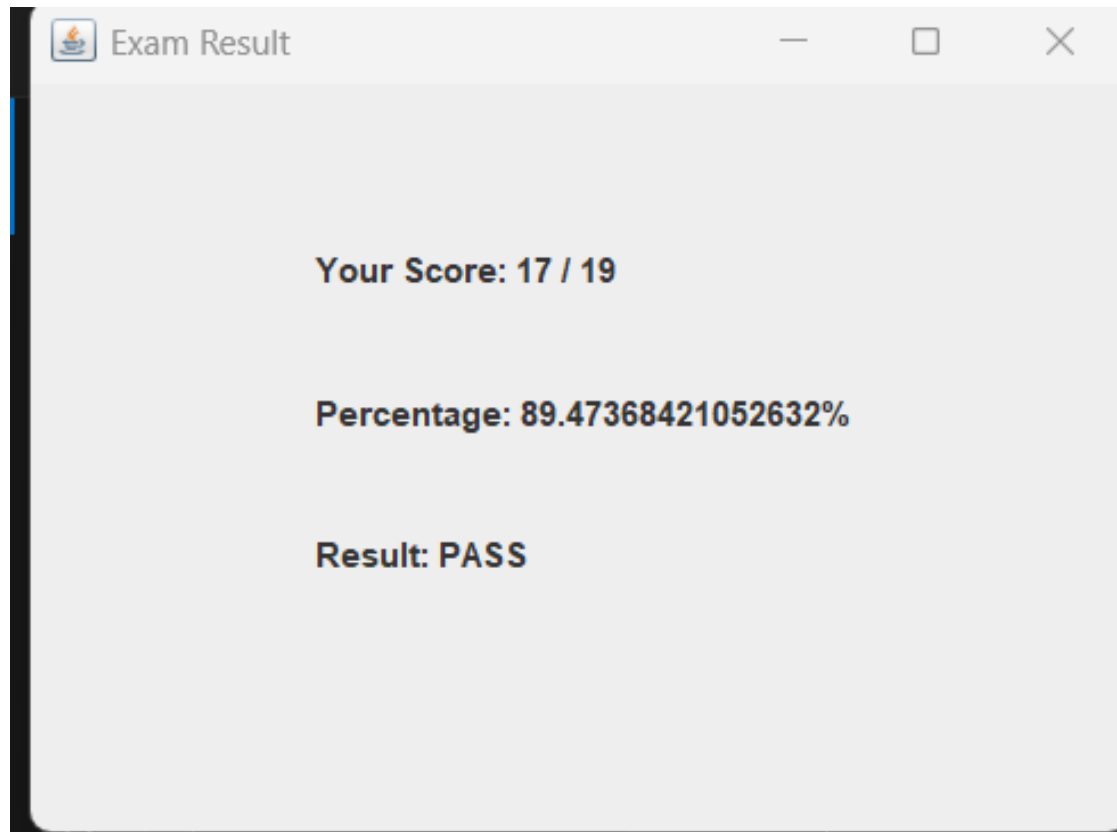


Figure 6.4: Result Dashboard

Chapter 7 – Conclusion and Future Scope

Conclusion

The **Online Examination System** project was successfully developed and tested using Java and MySQL.

This project improved knowledge in Java programming, database connectivity, GUI design, and exception handling.

Future Enhancements

- Cloud integration for online access
- Mobile application support
- Online payment gateway integration
- Advanced security and authentication
- Performance analysis and report generation

References

Students may include:

1. Java: The Complete Reference.
2. Thinking in Java.
3. MySQL Workbench.
4. Java Swing Documentation
5. JDBC API Documentation.
6. [Oracle Java Documentation](#)